# Experiment-4

**Student Name:** Pranav Chuahan
**UID:** 24BCY70004
**Branch:** BE CSE (cyber security )
**Section/Group:**24AIT_KRG-G2
**Semester:** 5th
**Date:** 29/07/2026
**Subject Name:** Computer Networks
**Subject Code:** 24CSH-337

## 1. Aim/Overview of the practical:

To implement and configure **Static Routing** using multiple routers in **Cisco Packet Tracer**, assign IP addresses to network devices, establish communication between different networks by manually configuring routing tables, and verify end-to-end connectivity using network testing commands such as **ping**.

## 2. Software/Tools Required:

- Cisco Packet Tracer
- Windows Operating System

## 3. Theory:

### A) What is Routing?

Routing is the process of selecting the best path for forwarding data packets from a source network to a destination network. When devices belong to different networks, routers use routing tables to determine the appropriate path for transmitting data.

### B) What is Static Routing?

Static Routing is a routing technique in which routes are manually configured by the network administrator. The router does not automatically learn network paths; instead, each route is entered manually using static route commands. Static routing is commonly used in small and stable networks where the network topology changes infrequently.

### C) Working of Static Routing

In static routing, every router maintains a manually configured routing table containing information about remote networks and the next-hop IP address or exit interface. When a data packet reaches the router, it checks its routing table and forwards the packet through the specified path. Since routes are manually configured, any change in the network requires the administrator to update the routing table accordingly.

### D) Advantages of Static Routing

- Simple to configure and understand for small networks.
- More secure because routing information is not exchanged between routers.
- Requires less CPU and memory since no routing protocol calculations are performed.
- Consumes less network bandwidth because no routing updates are transmitted.

### E) Disadvantages of Static Routing

- Every route must be configured manually.
- Difficult to manage and maintain in large networks.
- Does not automatically adapt to network changes.
- Requires manual updates whenever the network topology changes.

| Feature | Static Routing | Dynamic Routing |
|---|---|---|
| **Configuration** | Routes are configured manually by the administrator. | Routes are learned and updated automatically using routing protocols. |
| **Route Updates** | Manual updates are required whenever the network changes. | Routes are updated automatically when the network topology changes. |
| **Scalability** | Suitable for small networks. | Suitable for medium and large networks. |
| **CPU Usage** | Low, as no route calculations are performed. | Higher, due to continuous route calculations. |
| **Memory Usage** | Low memory requirement. | Higher memory requirement for maintaining routing tables. |
| **Bandwidth Usage** | Low, since no routing updates are exchanged. | Higher, because routing updates are periodically exchanged. |
| **Security** | More secure as routing information is not advertised. | Less secure due to routing information exchange. |
| **Maintenance** | Difficult to maintain in large networks. | Easier to maintain in large and changing networks. |
| **Network Changes** | Requires manual reconfiguration. | Automatically adapts to network changes. |
| **Best Suited For** | Small and stable networks. | Medium and large networks with frequent topology changes. |

## 4. Steps for Experiment/Practical:

### A) Static Routing using 4 PCs, 2 Routers and 2 Switches

Procedure / Steps:

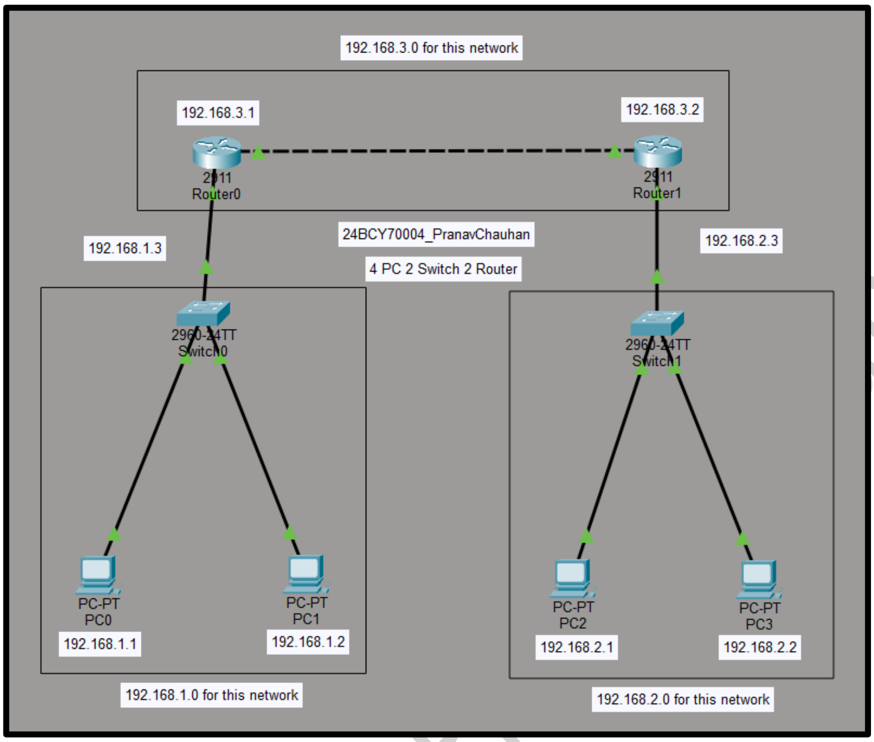
1. Open Cisco Packet Tracer and place 4 PCs, 2 Routers, and 2 Switches in the workspace.
2. Connect all devices using appropriate cables according to the network topology.
3. Assign IP addresses to all PCs and configure IP addresses on the router interfaces.
4. Configure the Default Gateway on each PC.
5. Configure Static Routes on both routers.
6. Verify the routing table using the command:
   **show ip route**
7. Test connectivity between the two PCs using the ping command.

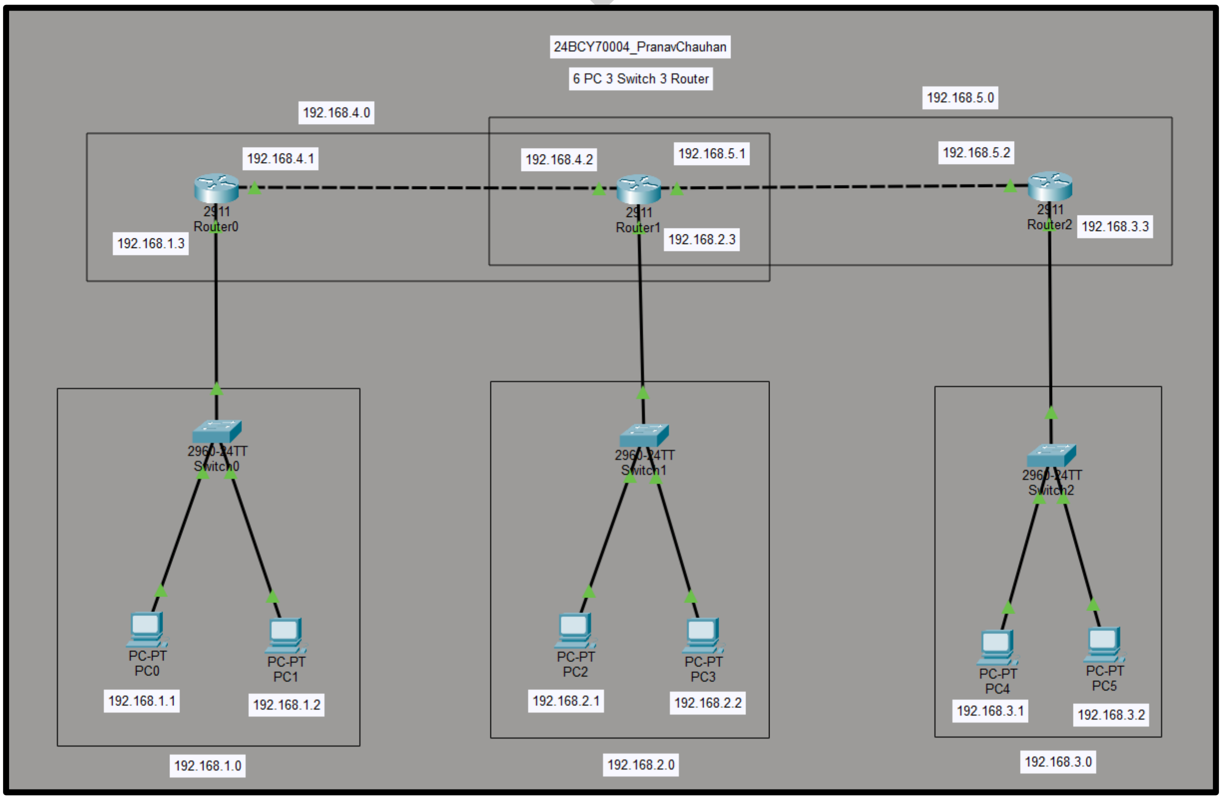
### B) Static Routing using 6 PCs, 3 Routers and 3 Switches

Procedure / Steps:

1. Open Cisco Packet Tracer.
2. Open Cisco Packet Tracer and place 6 PCs, 3 Routers, and 3 Switches in the workspace.
3. Connect all devices using appropriate cables according to the network topology.
4. Assign IP addresses to all PCs and configure IP addresses on the router interfaces.
5. Configure the Default Gateway on each PC.
6. Configure Static Routes on all three routers.
7. Verify the routing table using the command:
   **show ip route**
8. Test connectivity between PCs on different networks using the ping command.

## 5. Output:




24BCY70004_PranavChauhan

## 6. Result

The implementation of **Static Routing** using multiple routers was carried out successfully in **Cisco Packet Tracer**. IP addresses and static routes were configured manually on all routers, enabling successful communication between different networks. Network connectivity was verified using the **ping** command, and the routing tables were confirmed using the **show ip route** command. The experiment demonstrated the successful implementation and working of static routing.

## 7. Learning Outcomes

- Understood the concept and working of Static Routing in computer networks.
- Learned to configure static routes manually on routers using Cisco Packet Tracer.
- Gained practical experience in assigning IP addresses and configuring router interfaces.
- Verified communication between different networks using the ping command and show ip route.
- Developed the ability to design, configure, and troubleshoot small routed networks using static routing.